Fiscal Year:	FY 2008	**Task Last Updated:**	FY 01/30/2008
PI Name:	Sundaresan, Alamelu Ph.D.		
Project Title:	Targeting specific bone cell signaling pathways and immunosuppression in microgravity		
Division Name:	Human Research		
Program/Discipline:	HUMAN RESEARCH		
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Operational and clinical research		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	alamelu.sundaresan@tsu.edu	**Fax:**	FY
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Organization Name:	Texas Southern University		
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PI Web Page:			
City:	Houston	**State:**	TX
Zip Code:	77004	**Congressional District:**	18
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	UNSOL-JSC
Start Date:	01/01/2006	**End Date:**	12/31/2008
No. of Post Docs:	0	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA JSC
Contact Monitor:	Pellis, Neal	**Contact Phone:**	
Contact Email:	neal.r.pellis@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: Received NCE to 12/31/2008 per PI (12/07) NOTE: Received NCE to 12/31/2007 per PI (12/06)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:	NNJ06HA40G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The utility of the above project is apparent, in view of the high level of importance placed by NASA on understanding and developing countermeasures to the effects of microgravity exposure and consequent unloading-induced bone loss and immune suppression in astronauts. Both these projects could provide novel information about the effects of unloading on the complex cellular interactions between cells of the skeletal system and the potential use of pharmacological countermeasures such as statins and bisphosphonates. The signaling experiments would delineate target molecules which could be manipulated to maintain equilibrium even in unloaded conditions. Cellular and tissue level changes occur much earlier than whole organ changes, which can be addressed in both these projects, thereby providing a mechanistic understanding (cell signaling modulation within the bone matrix/immune cell complex) and basis for developing whole body countermeasures for alleviating bone loss and immune suppression from the space flight perspective. Both these projects will involve travel to national and international NASA related and basic science related symposia by the PI. Task Significance: The project will provide a mechanistic understanding (cell signaling modulation within the bone matrix/immune cell complex) and basis for developing whole body countermeasures for alleviating bone loss and immune suppression from the space flight perspective.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Both these projects could provide novel information about the effects of unloading on the complex cellular interactions between cells of the skeletal system and the potential use of pharmacological countermeasures such as statins and bisphosphonates.
Task Progress:	The project consists of two components: (1) Investigation of a three dimensional human bone tissue model as a physiological test bed for testing bone loss countermeasures and (2) Lymphocyte signaling and immune suppression in microgravity. This year the progress has been good, with two manuscripts submitted and under peer review — one to the “Annals of New York Academy of Sciences” and one to “Microgravity Science and Technology”. Investigation of this marker may provide valuable insight for future spaceflight relative to the cardiovascular system as a consequence of immune suppression. Relative to the budget, reagent kits for human astronaut samples for a cardiovascular biomarker are on back order until March 2008, and some money allocated for graduate students has been freed up by scholarships allotted to them. Specific bone signaling kits and immune kits have been ordered and on back order. These will be delivered in April.
Bibliography Type:	Description: (Last Updated: 07/02/2026)
Abstracts for Journals and Proceedings	Sundaresan A. "Regulation of Raf/MEK kinases in lymphocytes in microgravity-Responses to Cellular Stress: Invited talk." “Young Life Scientist” (Biochemical Society) conference, Glasgow, UK, July 7-10, 2007. “Young Life Scientist” (Biochemical Society) conference in Glasgow, UK., July 7-10, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Sundaresan A. "Research talk." UNCSFP_NAFP (NASA administrator’s Fellowship) conference, Cleveland, Ohio, July 16-19, 2007. Recent Research talk at the UNCSFP_NAFP (NASA administrator’s Fellowship) conference, Cleveland, Ohio, July 16-19, 2007. , Jul-2007
Abstracts for Journals and Proceedings	Sundaresan A, DuMond J, Singh K, Pellis NR. "Genetic signatures during physiological adaptation. Selected for oral presentation." Selected for oral presentation in the “Gene Expression Profile” session on Wednesday, September 5, 2007, European Low Gravity Research Association meeting, Florence, Italy, September 3-7, 2007. European Low Gravity Research Association meeting, September 3-7, 2007. , Sep-2007
Abstracts for Journals and Proceedings	Sundaresan A. "Cellular and genetic adaptation in low gravity environments (Bio Response in Space Environments). Invited talk." Invited talk at the Engineering Conferences International meeting in Bansko, Bulgaria, Oct 14-17, 2007. Engineering Conferences International meeting, Bansko, Bulgaria, Oct 14-17, 2007. , Oct-2007
Articles in Peer-reviewed Journals	Taga M, Yamauchi K, Odle J, Furian L, Sundaresan A, Ramesh GT, Pellis NR, Andrassy RJ, Kulkarni AD. "Melanoma growth and tumorigenicity in models of microgravity." Aviat Space Environ Med. 2006 Nov;77(11):1113-6. PMID: 17086762 , Nov-2006
Articles in Peer-reviewed Journals	Kaur C, Sivakumar V, Ang LS, Sundaresan A. "Hypoxic damage to the periventricular white matter in neonatal brain: role of vascular endothelial growth factor, nitric oxide and excitotoxicity." J Neurochem. 2006 Aug;98(4):1200-16. PMID: 16787408 , Aug-2006
Articles in Peer-reviewed Journals	Sundaresan A, Kulkarni AD, Yamauchi K, Pellis NR. "The role of nucleotides in augmentation of lymphocyte locomotion: Adaptational countermeasure development in microgravity analog environments." Microgravity Science and Technology 2006;18(3-4):247-9. , Nov-2006
Articles in Peer-reviewed Journals	Sundaresan A, Pellis NR. "Gene Regulation in modeled microgravity." Manuscript submitted to Annals of the NY Academy of Sciences, accepted December, 2007. , Dec-2007